

Kian Faizi

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Education

California Institute of Technology

PhD in Systems Biology

Pasadena, CA

Sep. 2021 – present

University of California, San Diego

BS in Molecular Biology, Minor in Mathematics

La Jolla, CA

Aug. 2017 – June 2021

Experience

California Institute of Technology

PhD Student, Dr. Rob Phillips

Pasadena, CA

Sep. 2021 – present

Harvard Medical School

Visiting Postgraduate Fellow, Dr. Pam Silver

Boston, MA

June 2022 – Aug. 2022

Salk Institute for Biological Studies

Lab Technician, Dr. Wolfgang Busch

La Jolla, CA

Nov. 2019 – Sep. 2021

Volunteer Research Assistant, Dr. Patrick Hsu

Nov. 2018 – Nov. 2019

Publications

7. **Kian Faizi**, Niko McCarty, and Sasha Targ. "Free-living chloroplasts: a preliminary investigation." *New Science* (2024). [newscience.org](https://www.newscience.org)
6. Rosalind Wenshan Pan, Tom Röschinger, **Kian Faizi**, and Rob Phillips. "Dissecting endogenous genetic circuits from first principles." *arXiv* (2024). [10.48550/arXiv.2401.15880](https://arxiv.org/abs/10.48550/arXiv.2401.15880)
5. Jingyi Wei, Peter Lotfy, **Kian Faizi**, Sara Baungaard, Emily Gibson, Eleanor Wang, Hannah Slabodkin, Emily Kinnaman, Sita Chandrasekaran, Hugo Kitano, Matthew G. Durrant, Connor V. Duffy, April Pawluk, Patrick D. Hsu, and Silvana Konermann. "Deep learning and CRISPR-Cas13d ortholog discovery for optimized RNA targeting." *Cell Systems* (2023). [10.1016/j.cels.2023.11.006](https://doi.org/10.1016/j.cels.2023.11.006)
4. Matthieu Pierre Platre, Preyanka Mehta, Zachary Halvorson, Ling Zhang, Lukas Brent, Matias Gleason F., **Kian Faizi**, Callum Goulding, and Wolfgang Busch. "Root Walker: an automated pipeline for large scale quantification of early root growth responses at high spatial and temporal resolution." *The Plant Journal* (2023). [10.1111/tpj.16493](https://doi.org/10.1111/tpj.16493)
3. **Kian Faizi**. "In Defense of Basic Science." *Caltech Letters* (2022). caltechletters.org
2. **Kian Faizi**. "Reversing 1.5 Billion Years of Evolution." *New Science* (2022). [10.56416/720qud](https://www.newscience.com/story/10.56416/720qud)
1. Illia Ziamtsov, **Kian Faizi**, and Saket Navlakha. "Branch-Pipe: Improving graph skeletonization around branch points in 3D point clouds." *Remote Sensing*. (2021). [10.3390/rs13193802](https://doi.org/10.3390/rs13193802)

Honors and Awards

Center for Evolutionary Science project grant (\$10k) | Caltech

Feb. 2024

New Science summer fellowship (\$33k) | New Science Research, Inc.

Mar. 2022

Halicioğlu Data Science Institute scholarship project award | UCSD

May 2021

Halicioğlu Data Science Institute scholarship (\$2.5k) | UCSD

Dec. 2019

Eureka! scholarship (\$5k) | UCSD

June 2019

Teaching

Physical Biology of the Cell teaching assistant (course site) MPI-CBG, Dresden	2024
Galápagos Expedition and DNA Bootcamp teaching assistant Caltech	2024
Evolution teaching assistant Caltech	2024
Current Research in Biology and Biological Engineering teaching assistant Caltech	2023, 2024
Physical Biology Bootcamp teaching assistant Caltech	2023, 2024
The Great Ideas of Biology head teaching assistant (course site) Caltech	2023, 2024
Principles of Biology teaching assistant Caltech	2022
Genetic Inquiry instructional assistant UCSD	2020

Mentorship, Leadership, and Service

Caltech GO-Outdoors leading field-based education for local K-12 classrooms (site)	2024 – present
Technical advisor for various climate-focused funding programs:	2023 – present
- Homeworld Collective Garden Grants: reviewed proposals on protein engineering for sustainability (\$1m awarded) (site) 50Y Manifest Climate Grants: evaluated “fast grants” for high-risk, high-reward climate biotech projects (\$1m awarded) (site) - Caltech Rocket Fund: prepared and presented due diligence reports on cleantech startups competing for non-dilutive funding (~\$800k awarded) (site)	
Research mentor for visiting high school students:	2023 – present
G. Milo: modeling freshwater eutrophication and algal blooms S. Bhattacharya, A. Bhate: mapping human impacts on global phosphorus cycling	
Caltech Rocket Fund editorial staff for monthly cleantech newsletter	2023 – present
Pasadena Unified Innovation Expo volunteer judge for K-8 science fair (site)	2023, 2024
Caltech Biology lead organizer for recruitment weekend	Jan. 2023
Caltech Annual Teaching Conference workshop facilitator for new TAs (site)	Sep. 2022
Caltech Accountability Partners Program peer mentor for prospective students (site)	Aug. 2022
Caltech Mycology Club member; helped establish indoor co-op mushroom farm	Dec. 2021 – present
UCSD Tritons Connect peer mentor for alumni	May 2021 – May 2022
UCSD Undergraduate Bioinformatics Club member	Nov. 2017 – June 2021